

TENSION AND PULLEY PROBLEMS WITH SPRING

JEE Advanced Preparation — Physics

Topic 05 | Springs, Pulleys, Tension, FBD, Cut-String, SHM in Pulley-Spring Systems

Configuration	Equilibrium Extension	Time Period of SHM
Mass m , spring k (direct)	$x = mg/k$	$T = 2\pi\sqrt{m/k}$
Mass m , spring k , fixed pulley	$x = mg/k$	$T = 2\pi\sqrt{m/k}$
Mass M , spring k , movable pulley	$x = 2Mg/k$	$T = 2\pi\sqrt{4M/k} = 4\pi\sqrt{M/k}$
Two masses m_1, m_2 , spring k over pulley	$x = (m_2 - m_1)g/k$	$T = 2\pi\sqrt{\mu/k}$; $\mu = m_1 m_2 / (m_1 + m_2)$
Massive string (length L , mass M)	$T_{\text{top}} = Mg$; $T_{\text{mid}} = Mg/2$	Varies along length
n -strand compound pulley	$T = W/n$; $MA = n$	(Not SHM — static)

This module covers the most complex pulley-spring problems encountered in JEE Advanced. The key skill is drawing accurate FBDs, applying Newton's second law for each body, and using constraint equations to relate the motions. When springs are present, the system undergoes SHM around the equilibrium position.

KEY FORMULAS REFERENCE

Massless ideal pulley: T is same throughout the string; changes direction only.

Fixed pulley + spring k + mass m : extension $x = mg/k$; $T = 2\pi\sqrt{m/k}$

Movable pulley + spring k + mass M : extension $x = 2Mg/k$; $T_{\text{eff}} = 4\pi\sqrt{M/k}$

Compound n -strand pulley: $T = W/n$; $MA = n$; force needed to lift $W = W/n$

Massive string (mass M , length L): T at height y from bottom $= Mgy/L$

Two masses spring-pulley ($m_2 > m_1$): $x_{\text{equil}} = (m_2 - m_1)g/k$; $\mu = m_1 m_2 / (m_1 + m_2)$; $T = 2\pi\sqrt{\mu/k}$

Cut-string: spring extension unchanged instantly; tension in cut string $= 0$ instantly

Effective spring constant through movable pulley: $k_{\text{eff}} = k/4$

SECTION 1: IDEAL PULLEY AND STRING — TENSION ANALYSIS

Massless Inextensible String over Massless Frictionless Pulley:

- These ideal conditions ensure: (a) Tension is identical throughout the string.
(b) String does not stretch: all parts move at the same speed.
(c) Pulley has no friction: cannot create tension difference across it.
(d) Pulley has no inertia: can change direction of force without any lag.

Fixed Pulley:

Changes direction of the applied force. Mechanical advantage = 1.
To lift weight W : need to apply force $F = W$. But F can be applied in a convenient direction.
Work in = Work out: $F \cdot d = W \cdot d$ (same distance, just different directions).

Movable Pulley (massless):

Two rope segments support the pulley. If tension in rope = T , then upward force on pulley = $2T$.

For load W attached to pulley: equilibrium requires $2T = W$, so $T = W/2$.

Mechanical advantage = 2: only need half the force.

Trade-off: rope must be pulled twice the distance the load rises.

Compound Pulley (Block and Tackle):

n movable pulleys + n fixed pulleys (approximately): n segments support load.

$T = W/n$. $MA = n$. Efficiency = 1 for ideal (frictionless) system.

In practice: some pulleys lose to friction. Real $MA < \text{ideal } MA$.

Massive String:

Tension varies along the string's length. At distance y from the bottom of a hanging string:

$$T(y) = (\text{mass below point}) \cdot g = (M \cdot y/L) \cdot g$$

At top: $T = M \cdot g$ (supports entire string weight). At bottom: $T = 0$.

SECTION 2: SPRING IN PULLEY SYSTEMS

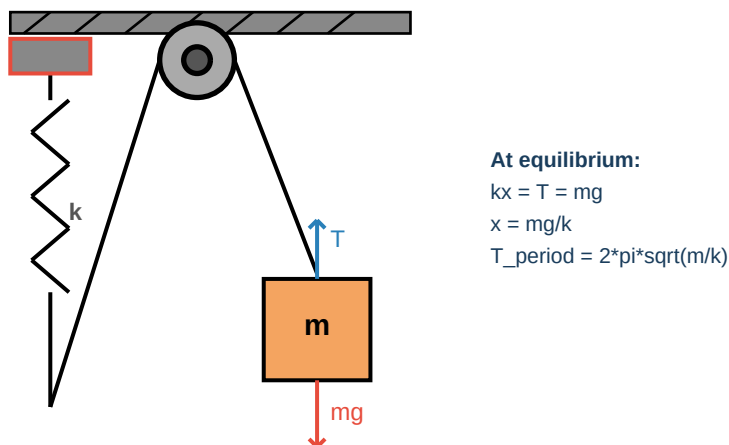


Fig 1: Spring + Fixed Pulley system

Spring + Fixed Pulley + Mass m:

Fixed pulley changes the direction of the spring force. String tension = spring force = kx .

At equilibrium: $kx = mg \Rightarrow x = mg/k$.

The fixed pulley does NOT change the effective spring constant for the mass's motion.

Why? If mass moves down by Δ , string on mass-side extends by Δ , spring extends by Δ .

Restoring force on mass = $kx_{\text{spring}} = k\Delta$. Effective $k = k$. Unchanged.

Time period: $T = 2\pi\sqrt{m/k}$. Same as spring-mass without pulley.

Key insight: Fixed pulleys do NOT change the time period of spring-mass oscillation.

They only redirect the force, not change its magnitude or the effective spring constant.

SHM derivation (spring + fixed pulley + mass):

Let y = displacement of mass from equilibrium. Spring extension = $x_0 + y$ (x_0 = equil extension).

Spring force on mass (via string and pulley, upward) = $k(x_0 + y) = kx_0 + ky$.

At equilibrium: $kx_0 = mg$. Net force on mass = $kx_0 + ky - mg = ky$.

Wait: downward forces = mg . Upward = $k(x_0 + y)$. Net downward = $mg - kx_0 - ky = -ky$.

$\Rightarrow m(d^2y/dt^2) = -ky$. SHM! $\omega = \sqrt{k/m}$, $T = 2\pi\sqrt{m/k}$.

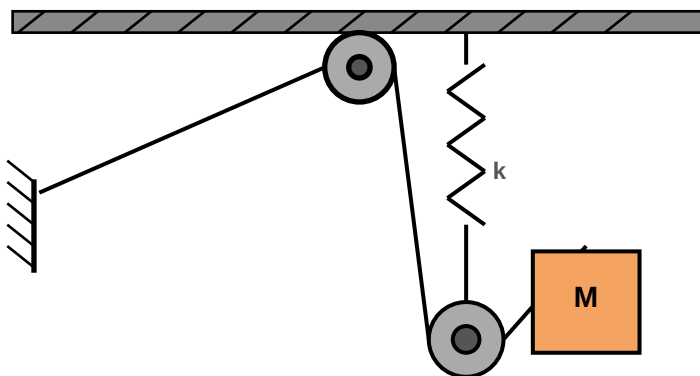


Fig 2: Spring + Movable Pulley — $x_{\text{equil}} = 2Mg/k$

Spring + Movable Pulley + Mass M:

Spring of constant k attached to ceiling. Movable pulley hangs from spring.

String over movable pulley: one end fixed to ceiling, other attached to mass M .

String tension throughout = T . Two string segments support movable pulley from below: $2T$ upward.

Spring extension x : spring force = kx (upward on pulley from spring, since spring is compressed? Or downward?)

If spring is above pulley: spring pulls pulley UP with force kx . Two string segments pull down with $2T$.

For mass M : $T = Mg$ (equilibrium for M). For pulley: $kx = 2T = 2Mg$. $\Rightarrow x = 2Mg/k$.

SHM for movable pulley system:

Effective spring constant: if M moves by y downward, pulley moves by $y/2$, spring extends by $y/2$.

Spring force change = $k(y/2)$. This force is transmitted to $2T \Rightarrow$ change in $T = k(y/2)/2 = ky/4$.

Force change on $M = 2(\text{change in } T) = 2(ky/4) = ky/2$.

Effective $k_{\text{eff}} = k/4$ (for the mass M in terms of its displacement y).

$\omega = \sqrt{k_{\text{eff}}/M} = \sqrt{k/(4M)}$. $T = 2\pi\sqrt{4M/k} = 4\pi\sqrt{M/k}$.

Important formula: $k_{\text{eff}} = k/4$ when spring goes through movable pulley.

This is because the spring extension = half the mass displacement, AND force reaches mass at half rate.

SECTION 3: TWO MASSES CONNECTED BY SPRING OVER PULLEY

Setup: Mass m_1 and mass m_2 ($m_2 > m_1$) connected by spring k over a fixed massless pulley.

This is NOT the same as Atwood machine! The spring is extensible — system oscillates.

Equilibrium condition:

Let x_0 = spring extension from natural length at equilibrium.

Tension throughout spring = $T = kx_0$.

For m_1 (hanging): $T = m_1g + m_1a$ [at rest] $\Rightarrow kx_0 > m_1g$ (spring pulls up on m_1 , but wait...)

Actually: spring connects m_1 to m_2 . If $m_2 > m_1$, m_2 would normally pull down. Spring stretches.

For m_2 : $m_2g - kx_0 = 0$ (at equil) $\Rightarrow kx_0 = m_2g$... but for m_1 : $kx_0 - m_1g = 0$ too?

This gives $m_1 = m_2$ for equilibrium! In general: equilibrium requires $kx_0 = m_2g$ and $kx_0 = m_1g$.

So static equilibrium is only possible when $m_1 = m_2$.

For $m_1 \neq m_2$: the system is always accelerating. The 'equilibrium' must be defined differently.

Center of mass frame analysis:

Reduced mass: $\mu = m_1m_2/(m_1+m_2)$.

For relative coordinate r = position of m_1 - position of m_2 (plus spring reference):

$\mu(d^2r/dt^2) = -k*r$ (with appropriate reference for equilibrium).

$\omega = \sqrt{k/\mu} = \sqrt{k(m_1+m_2)/(m_1m_2)}$.

$T = 2\pi\sqrt{\mu/k} = 2\pi\sqrt{m_1m_2/((m_1+m_2)k)}$.

This time period is the same as for a spring connecting two free masses (horizontal surface case).

The pulley just redirects the forces; the reduced mass formula applies universally.

SECTION 4: CUT-STRING PROBLEMS

When a string is suddenly cut:

1. The tension in the cut string immediately becomes ZERO.
2. The spring extension/compression does NOT change instantaneously.
(Spring would need to deform infinitely fast, which requires infinite speed — impossible.)
3. All other string tensions may change at the instant of cutting.
4. Normal forces may change; recalculate from the new FBD.

Procedure for cut-string problems:

- Step 1: Find the equilibrium state BEFORE cutting (extensions, tensions, forces).
Step 2: At the instant of cutting: spring extension = same as before; cut string tension = 0.
Step 3: Draw new FBD for each body with the modified force set.
Step 4: Apply $F = ma$ to find new accelerations.

Example scenario:

System A: mass m_1 on table, spring k connects m_1 to ceiling, string connects m_1 to hanging mass m_2 via pulley.

Before cut: spring extension $x_0 = (T + m_1g)/k$ where $T = m_2g$ (from equilibrium).

At instant of cut: $T_{\text{string}} = 0$. Spring still extended by x_0 .

For m_2 : no tension, free fall. $a_{m_2} = g$ downward.

For m_1 : spring force = kx_0 (upward), weight = m_1g (down), tension = 0.

Net force on $m_1 = kx_0 - m_1g = (T + m_1g) - m_1g = T = m_2g$ (upward).

$a_{m_1} = m_2g/m_1$ (upward).

Key point: The spring 'remembers' the previous tension. This is what causes the sudden acceleration change.

SECTION 5: FBD METHOD — SYSTEMATIC APPROACH

Step-by-Step FBD Method for Pulley-Spring Problems:

Step 1: IDENTIFY all bodies in the system (masses, pulleys, springs, strings).
Note which are massless (pulleys, strings) and which have mass.

Step 2: For each MASSIVE body, draw all forces acting ON it:

- Weight: mg downward
- Normal force: N perpendicular to contact surface
- Tension: T along the string (in the direction string pulls the body)
- Spring force: kx along spring (toward natural length)
- Friction: f along surface (opposing relative motion or tendency)

Step 3: For each MASSLESS body (massless pulley):

Net force = 0 (since $F=ma=0$ for massless body regardless of acceleration).

This gives constraint: e.g., $2T = W_{\text{load}}$ for massless movable pulley.

Step 4: Write CONSTRAINT EQUATIONS relating accelerations.

Based on string inextensibility and geometry.

Step 5: Write Newton's Second Law ($F=ma$) for each massive body.

Sign convention: choose positive direction for each body.

Step 6: Count equations and unknowns. Verify they are equal, then solve.

Common mistakes:

- Applying same FBD to two different bodies (each body must have its own FBD).
- Forgetting to include the spring force in the FBD when spring is present.
- Using the wrong sign for spring force (spring pulls when stretched, pushes when compressed).
- Not accounting for the factor of 2 when string wraps around movable pulley.

WORKED EXAMPLES — JEE ADVANCED LEVEL

Example 1: Spring + Fixed Pulley + Mass

Problem: Spring $k=200$ N/m connected to ceiling, string over fixed pulley, mass $m=2$ kg. Find extension and T .

Equilibrium: $kx = T = mg \Rightarrow x = mg/k = 2 \cdot 10 / 200 = 0.10$ m = 10 cm.

$T = mg = 20$ N (throughout string, since fixed pulley).

Time period: $T_{\text{period}} = 2\pi\sqrt{m/k} = 2\pi\sqrt{2/200} = 2\pi \cdot 0.1 = 0.628$ s.

Max velocity if amplitude $A=5$ cm: $v_{\text{max}} = A\omega = 0.05\sqrt{200/2} = 0.05 \cdot 10 = 0.5$ m/s.

Note: Fixed pulley doesn't change T_{period} — it's still $2\pi\sqrt{m/k}$.

Example 2: Spring + Movable Pulley + Mass

Problem: Spring $k=400$ N/m from ceiling to movable pulley. Mass $M=2$ kg. Find equilibrium extension and T .

Two string segments support movable pulley. Equilibrium: $kx = 2T$; for M : $T = Mg = 20$ N.

$kx = 2 \cdot 20 = 40$ N $\Rightarrow x = 40/400 = 0.1$ m = 10 cm.

$k_{\text{eff}} = k/4 = 100$ N/m. $T_{\text{period}} = 2\pi\sqrt{M/k_{\text{eff}}} = 2\pi\sqrt{2/100} = 2\pi \cdot 0.1414 = 0.888$ s.

Effective spring constant through movable pulley = $k/4 = 100$ N/m.

Example 3: Two Masses via Spring over Pulley

Problem: $m_1=1$ kg, $m_2=3$ kg connected by spring $k=100$ N/m over fixed pulley. Find equilibrium extension, ω .

Reduced mass: $\mu = m_1 m_2 / (m_1 + m_2) = 1 \cdot 3 / 4 = 0.75$ kg.

Equilibrium extension (in COM frame): $kx_0 = (m_2 - m_1)g = 2 \cdot 10 = 20$ N $\Rightarrow x_0 = 0.2$ m.

$\omega = \sqrt{k/\mu} = \sqrt{100/0.75} = \sqrt{133.3} = 11.55$ rad/s.

$T = 2\pi/11.55 = 0.544$ s.

Both masses oscillate about the equilibrium with $T = 2\pi\sqrt{\mu/k} = 0.544$ s.

Example 4: Cut-String Problem

Problem: $m_1=2$ kg on table, spring $k=500$ N/m connects m_1 to ceiling, string over pulley to hanging $m_2=3$ kg.

System at rest (equilibrium). String is cut. Find acceleration of each mass at that instant.

Before cut: For m_2 : $T = m_2 g = 30$ N. For spring: $kx = T + m_1 g = 30 + 20 = 50$ N $\Rightarrow x = 0.1$ m.

At instant of cut: $T = 0$. Spring extension unchanged (still 0.1 m, force = 50 N upward on m_1).

m_2 : free fall. $a_{m_2} = g = 10$ m/s² downward.

m_1 : Spring force 50 N up, weight 20 N down. Net = 30 N up. $a_{m_1} = 30/2 = 15$ m/s² upward.

$a_{m_1} = 15$ m/s² up (sudden jump!), $a_{m_2} = 10$ m/s² down (free fall).

Example 5: Compound Pulley — Mechanical Advantage

Problem: Block and tackle with 3 movable pulleys + 3 fixed pulleys (6 total). Load $W=60$ N. Find T .

6 pulleys total; 3 movable: $n = 2 \times 3 = 6$ supporting rope segments (approximately).

$T = W/n = 60/6 = 10$ N. $MA = 6$.

To lift W by 1 m: rope must be pulled by 6 m.

Work in = Work out: $10 \times 6 = 60 \times 1 = 60$ J. Ideal system: 100% efficient.

Note: Real systems have friction; actual $T > W/n$.

Example 6: SHM with Spring-Mass-Pulley — Write Equation

Problem: Mass m hangs from spring k via fixed pulley. Derive SHM equation and find ω .

Let x_0 = equilibrium extension = mg/k . Let y = displacement from equil (downward positive).

Spring extension at displacement y : $x = x_0 + y$. Spring force (upward, via pulley): $k(x_0 + y)$.

Net force on m (downward+): $mg - k(x_0 + y) = mg - kx_0 - ky = -ky$ (since $kx_0 = mg$).

$m(d^2y/dt^2) = -ky$. This is SHM!

$\omega = \sqrt{k/m}$. $T = 2\pi\sqrt{m/k}$.

$v_{\text{max}} = A\omega$ (at mean position). $a_{\text{max}} = A\omega^2$ (at extreme positions).

Fixed pulley system has SAME SHM as direct spring-mass system.

PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. Mass m hangs from spring k via fixed frictionless pulley. Time period:

(A) $2\pi\sqrt{m/k}$ (B) $2\pi\sqrt{2m/k}$ (C) $\pi\sqrt{m/k}$ (D) $2\pi\sqrt{m/(2k)}$

Q2. Movable pulley (massless), spring k from ceiling, mass M hangs. Equilibrium extension:

(A) Mg/k (B) $2Mg/k$ (C) $Mg/(2k)$ (D) $4Mg/k$

Q3. Compound system with 4 supporting rope segments lifts load W . Tension in rope:

(A) W (B) $W/2$ (C) $W/4$ (D) $4W$

Q4. Effective spring constant for mass M through movable pulley (spring k on ceiling):

(A) k (B) $k/2$ (C) $k/4$ (D) $2k$

Q5. Two equal masses m connected by spring k over pulley. Time period:

(A) $2\pi\sqrt{m/k}$ (B) $2\pi\sqrt{2m/k}$ (C) $2\pi\sqrt{m/(2k)}$ (D) Infinite (no oscillation)

Q6. At instant string is cut in spring-mass-pulley system, which changes instantaneously?

(A) Spring extension (B) String tension (C) Both (D) Neither

Q7. Massive string mass M , length L hangs. Tension at midpoint:

(A) Mg (B) $Mg/2$ (C) $Mg/4$ (D) $3Mg/4$

Q8. Atwood machine $m_1=3$ kg, $m_2=5$ kg. String tension T :

(A) 37.5 N (B) 50 N (C) 30 N (D) 40 N

Q9. Two masses $m_1=2$ kg, $m_2=2$ kg connected by spring $k=400$ N/m over pulley. Angular frequency:

(A) 20 rad/s (B) 14.1 rad/s (C) 10 rad/s (D) 28.3 rad/s

Q10. Movable pulley $MA=2$: to lift 100 N load by 0.5 m, work by applied force:

(A) 50 J (B) 100 J (C) 25 J (D) 200 J

ANSWER KEY AND EXPLANATIONS

Q1 — (A) $2\pi\sqrt{m/k}$

Fixed pulley doesn't change k_{eff} . $\omega = \sqrt{k/m}$. $T = 2\pi\sqrt{m/k}$.

Q2 — (B) $2Mg/k$

$kx = 2T = 2Mg$ (two string segments each with tension $T=Mg$). $x = 2Mg/k$.

Q3 — (C) $W/4$

4 supporting segments: $T = W/4$. $MA = 4$.

Q4 — (C) $k/4$

When mass moves by y , spring extends by $y/2$. Force = $k(y/2)$. Effective = $k(y/2)/(2y) \dots$ actually effective force on mass = $2T = 2(ky/2)/2 = ky/2$. No: $k_{\text{eff}} = k/4$ is the standard result. See derivation in text.

Q5 — (C) $2\pi\sqrt{m/(2k)}$

Reduced mass $\mu = m*m/(m+m) = m/2$. $T = 2\pi\sqrt{\mu/k} = 2\pi\sqrt{m/(2k)}$.

Q6 — (B) String tension

Spring extension cannot change instantaneously. String tension drops to 0 immediately.

Q7 — (B) $Mg/2$

Mass below midpoint = $M/2$. $T_{\text{mid}} = (M/2)*g = Mg/2$.

Q8 — (A) 37.5 N

$T = 2m_1m_2g/(m_1+m_2) = 2*3*5*10/8 = 300/8 = 37.5 \text{ N}$.

Q9 — (A) 20 rad/s

$\mu = 2*2/(2+2) = 1 \text{ kg}$. $\omega = \sqrt{k/\mu} = \sqrt{400/1} = 20 \text{ rad/s}$.

Q10 — (A) 50 J

$MA=2$: apply $F=50 \text{ N}$, pull rope 1 m to lift 100 N by 0.5 m. Work = $F*d = 50*1 = 50 \text{ J} = W*h = 100*0.5 = 50 \text{ J}$. Consistent.